# RONIT RAJIB PATTNAIK

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## EDUCATION

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|---|---|---|---|
| B.Tech, Mechanical Engineering | ITER, Siksha 'O' Anusandhan, Bhubaneswar, Odisha | 9.46 | 2023-2027 |
| Class XII | ODM Public School, Bhubaneswar, Odisha | 87% | 2022-2023 |
| Class X | ODM Public School, Bhubaneswar, Odisha | 87% | 2020-2021 |

## SKILLS

- **Design & Analysis:** SolidWorks, FEA & CFD
- **Programming:** Python Scripting for automation in CAD
- **Data Analysis:** MySQL, Python, Power BI dashboards.
- **Reinforcement Learning:** MuJoCo, SAC for simulation

## INTERNSHIP

**Product & Growth Intern | Fastpeer AI Corps | South Plainfield, New Jersey, USA (Remote)** **March 2026 - June 2026**

- Centralized documentation and reporting workflows across 4 AI products and 10+ member of engineering team and replaced fragmented communication channels.
- Generated 15K+ organic impressions, structuring lead-acquisition pipelines across 4 U.S. states (NY, NJ, PA, CT), and maintaining a CRM of 300+ qualified prospects.
- Synthesized stakeholder and user feedback from multiple product demonstrations, influencing 25+ feature enhancements and product decisions across web and mobile AI platforms.

**Undergraduate Research Intern | ITER, SOA | Bhubaneswar, Odisha, India (On-site)** **May 2025 - Dec 2025**

- Modeled non-premixed syn-gas/air combustion in a round-jet burner as a 2D axisymmetric CFD simulation
- Found that raising the syn-gas inlet temperature from 300 K to 500 K increased peak flame temperature while cutting jet-tip velocity from 75 m/s to 65 m/s (about 13%)
- Showed that inlet temperature has an insignificant effect on NOx formation, so preheating raises flame temperature without an emissions penalty

**Robotics and Manufacturing Intern | Fx UAV | Bhubaneswar, Odisha, India (On-site)** **Dec 2024 - Feb 2025**

- Worked with 3 flight-controller platforms (KK2.1.5, MiniPix, and Pixhawk) to assemble and configure UAV systems.
- Utilized ArduPilot and Mission Planner for flight-controller configuration, sensor calibration, parameter tuning, flight-mode setup, and pre-flight verification.
- Integrated receiver–transmitter communication, performed ESC calibration, and validated channel mapping.

## PATENTS & PUBLICATIONS

**Indian Patent Application (Published):** Modular Robotic Hand, Application No. 202631034393, published in the Indian Patent Journal (Vol. 13/2026, 27 Mar 2026).

**Indian Design Application (Granted)**: Modular Robotic Hand, Design Application No. 499872-001.

**Research paper**: 3rd author, paper on "Effect of Inlet Temperature of Jet Burner Using Syn-Gas", submitted for publication (2026)

## PROJECTS

**Low-Cost Prosthetic Hand Model | *Doc Link*, *Visualize*** **June 2 - Dec 2 2025**

- Led end-to-end development of a low-cost prosthetic hand, identifying affordability and accessibility as key problem areas and translating them into a functional product solution throgh 5 interations.
- FEA-validated under 68 N grip-force loading, achieving a factor of safety of 3.11 with a maximum deformation of 0.532 mm.

**Reinforcement Learning for Dexterous Hand Manipulation | *Project Link*** **July 17 - Aug 25, 2026**

- Developed a MuJoCo-based reinforcement learning pipeline using behavioral cloning and SAC for continuous-control hand manipulation with multi-stage grasp/lift reward function.
- Processed and refined 9 behavioral demonstrations, identifying data-quality and episode-termination issues through iterative training and debugging.

## EXTRA CURRICULAR

**Public Speaking**: 2x State level speaker at Odisha State Legislature Assembly, top 5 among 200+ participants in Viksit Bharat Youth Parliament 2025, 2026.

**Innovation Challenges:** 3x Ideathon (XIM ICAIET 2025, Startup Odisha Hack the Future 2025, Synergy INNOVISION 2026)

**Clubs**: Founding member in GFG, Core Operations team member